

Domain 3 – Cloud Technology & Services (34%) – Complete Topic List

3.1 Methods of Deploying and Operating in AWS

Knowledge:

- Different ways of provisioning & operating in AWS Cloud
- Ways to access AWS services
- Cloud deployment models: Cloud, Hybrid, On-premises
- Connectivity options: AWS VPN, AWS Direct Connect, public internet

Skills:

- Choosing between:
 - Programmatic access (APIs, SDKs, CLI)
 - AWS Management Console
 - Infrastructure as Code (IaC)
 - Deciding one-time vs repeatable operations
 - Identifying deployment models (cloud, hybrid, on-prem)
 - Identifying connectivity options (VPN, Direct Connect, public internet)
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3.2 AWS Global Infrastructure

Knowledge:

- AWS Regions, Availability Zones (AZs), Edge Locations
- High availability
- Use of multiple Regions
- Benefits of edge locations
- AWS Wavelength Zones and AWS Local Zones

Skills:

- Relationship between Regions, AZs, and Edge Locations
 - Achieving high availability with multiple AZs
 - Recognizing that AZs **do not share single points of failure**
 - When to use multiple Regions: disaster recovery, business continuity, low latency, data sovereignty
 - Benefits of Edge Locations: CloudFront, Global Accelerator
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3.3 AWS Compute Services

Knowledge:

- AWS Compute Services

Skills:

- Appropriate use of EC2 instance types: compute-optimized, storage-optimized
 - Appropriate container options: Amazon ECS, Amazon EKS
 - Appropriate serverless options: AWS Fargate, Lambda
 - Auto Scaling provides elasticity
 - Purpose of load balancers
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3.4 AWS Database Services

Knowledge:

- AWS Database Services
- Database migration

Skills:

- Decide when to use EC2-hosted vs AWS-managed databases
 - Identify relational databases: Amazon RDS, Amazon Aurora
 - Identify NoSQL databases: DynamoDB
 - Identify memory-based databases: ElastiCache
 - Identify database migration tools: AWS DMS, AWS SCT
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3.5 AWS Network Services

Knowledge:

- AWS Network Services

Skills:

- Components of a VPC: subnets, gateways
 - Security in VPC: Network ACLs, Security Groups
 - Purpose of Amazon Route 53
 - Edge services: CloudFront, Global Accelerator
 - Network connectivity options: AWS VPN, Direct Connect
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3.6 AWS Storage Services

Knowledge:

- AWS Storage Services

Skills:

- Uses for object storage: S3
 - Differences in S3 storage classes: Standard, IA, Glacier, Deep Archive
 - Block storage: Amazon EBS, instance store
 - File services: Amazon EFS, Amazon FSx
 - Cached file systems: AWS Storage Gateway
 - Lifecycle policies use cases
 - AWS Backup use cases
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3.7 AWS Artificial Intelligence / Machine Learning and Analytics Services

Knowledge:

- AWS AI/ML services: SageMaker, Lex, Kendra, etc.
- AWS Analytics services: Athena, Kinesis, Glue, QuickSight

Skills:

- Understanding AI/ML services and their tasks
 - Identifying analytics services for data analysis
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3.8 Other AWS Services

Knowledge:

- **Application Integration:** EventBridge, SNS, SQS
- **Business Applications:** Amazon Connect, SES
- **Customer Engagement:** AWS Activate, AWS IQ, AWS Managed Services, AWS Support
- **Developer Tools:** AppConfig, Cloud9, CloudShell, CodeArtifact, CodeBuild, CodeCommit, CodeDeploy, CodePipeline, CodeStar, X-Ray
- **End-User Computing:** AppStream 2.0, WorkSpaces, WorkSpaces Web
- **Frontend Web & Mobile:** Amplify, AppSync
- **IoT:** AWS IoT Core, AWS IoT Greengrass

Skills:

- Choose correct service for messaging, alerts, notifications
- Choose service for business applications
- Choose service for customer support
- Identify tools to develop, deploy, troubleshoot apps

- Identify services for end-user VM output
- Identify services for frontend & mobile apps
- Identify services for managing IoT devices